

Derived Solid Realization

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Summary

Pass 77 corrects the target for the all-prime phantom

$$\epsilon = \widehat{\mathbb{Z}}/\mathbb{Z}.$$

The ordinary LCA route is too small: the dense diagonal image kills continuous characters, so the LCA dual of ϵ is zero. The useful realization is instead derived/solid. In that setting the profinite completion contributes a degree shift:

$$\widehat{\mathbb{Z}}^* \simeq (\mathbb{Q}/\mathbb{Z})[-1],$$

and the phantom survives as a derived object rather than as an ordinary exact LCA quotient.

Main Claims

- **LCA no-go.** Finite shadows show that the annihilator of the dense diagonal image is trivial at every checked lcm stage.
- **Solid degree shift.** For each finite stage $\mathbb{Z}/N$, Hom into $\mathbb{Z}$ vanishes while $\text{Ext}^1(\mathbb{Z}/N, \mathbb{Z})$ has order N .
- **Signed boundary law.** The finite boundary $d_S : \mathbb{Z}^S \rightarrow \mathbb{Z}^{|S|-1}$ has diagonal kernel, is surjective, and dualizes as $D(d_S) = -d_S^T$ with $D^2 = \text{id}$.

Machine Verification

`code/scripts/check-pass77.py` produced `artifacts/reports/pass77-derived-solid-realization-check.json` with overall PASS. It checks the LCA annihilator no-go, the solid degree-one dual tower, and the signed dual boundary law for support sets of sizes 2 through 6.

Next Use

This pass fixes the ambient category for the later ϵ analysis: subsequent symplectic, metaplectic, and automorphic statements must be read in the derived/solid target, not in ordinary LCA groups.